

Caifen Liu

PhD Candidate in Epidemiology and Biostatistics

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RESEARCH PROFILE AND INTERESTS

I am interested in developing and applying causal inference methods to observational health data to reduce bias and inform clinical decisions.

Methodology: causal estimands; selection bias; inverse-probability weighting; survival and longitudinal analysis; simulation and sensitivity analysis; missing data; target trial emulation.

Applications: linked electronic health records, registries, and cohorts; vaccine effectiveness; disease severity; pharmacoepidemiology.

EDUCATION

The University of Hong Kong Hong Kong, China

PhD in Epidemiology and Biostatistics Sep 2022–Sep 2026 (expected)

Thesis: *Estimation and Interpretation of COVID-19 Severity and Vaccine Effectiveness*.

Supervisor: [Prof. Ben Cowling](#).

Southern University of Science and Technology Shenzhen, China

MSc in Statistics Sep 2018–Jun 2020

Supervisor: [Prof. Guo-Liang Tian](#).

Hunan University Changsha, China

BE in Materials Science and Engineering Sep 2014–Jun 2018

DOCTORAL RESEARCH

The University of Hong Kong *School of Public Health* 2022–Present

Causal estimands and selection bias in test-negative designs

Investigated selection bias in vaccine-effectiveness estimates using causal estimands, DAGs, analytical derivations, Monte Carlo simulation, and inverse-probability weighting.

COVID-19 severity and in-hospital progression

Analysed linked surveillance and electronic health records (about 2.8 million cases; 120,000 hospitalizations) using Bayesian models, Aalen–Johansen estimators, and Cox models.

Hospital pressure, admission criteria, and fatality risk

Examined admission severity, hospital load, and mortality using generalized additive models, marginal standardization, and descriptive decomposition.

Hospital-based test-negative design estimates (*ongoing*)

Simulating vaccine-effectiveness estimates under alternative selection mechanisms, with randomized-trial and adjusted-cohort benchmarks.

Target trial protocol for hospital admission policies (*design exercise*)

Specified eligibility, time zero, and analysis for severity-based admission strategies; assessed feasibility limitations from missing pre-admission measurements and interference.

PUBLICATIONS AND MANUSCRIPTS

Peer-Reviewed Publications

†Co-first author

1. **Liu, C.**, Okoli, G. N., Chen, R., Sullivan, S. G., and Cowling, B. J. (2026). SARS-CoV-2 vaccination and attenuation of breakthrough infection severity: A systematic global review and meta-analysis. *Clinical Infectious Diseases*, [ciag346](#).
2. Du, Z., **Liu, C.**[†], Bai, Y., Wang, L., Lim, W. W., Lau, E. H. Y., and Cowling, B. J. (2024). Predicting efficacies of fractional doses of vaccines by using neutralizing antibody levels: Systematic review and meta-analysis. *JMIR Public Health and Surveillance*, 10, [e49812](#).
3. Du, Z., **Liu, C.**[†], Wang, C., Xu, L., Xu, M., Wang, L., Bai, Y., Xu, X., Lau, E. H. Y., Wu, P., and Cowling, B. J. (2022). Reproduction numbers of severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) variants: A systematic review and meta-analysis. *Clinical Infectious Diseases*, 75(1), e293–e295.
4. Du, Z., Wang, C., **Liu, C.**[†], Bai, Y., Pei, S., Adam, D. C., Wang, L., Wu, P., Lau, E. H. Y., and Cowling, B. J. (2022). Systematic review and meta-analyses of superspreading of SARS-CoV-2 infections. *Transboundary and Emerging Diseases*, 69(5).
5. **Liu, C.**[†], Xu, L., Bai, Y., Xu, X., Lau, E. H. Y., Cowling, B. J., and Du, Z. (2022). Local surveillance of the COVID-19 outbreak. *Frontiers in Physics*, 10, 824369.

Working Papers

1. **Liu, C.**, Murphy, C., Okoli, G. N., Sullivan, S. G., and Cowling, B. J. *Vaccine-induced severity attenuation and bias in test-negative estimates of vaccine effectiveness*. Manuscript in preparation.
2. **Liu, C.**, Cheung, J. K., Lin, Y., Wu, P., Wong, J. Y., and Cowling, B. J. *Hospital pressure, admission selection, and COVID-19 hospital fatality risk across epidemic periods in Hong Kong: a retrospective cohort study*. Manuscript in preparation.

SELECTED PRESENTATIONS

EPIDEMICS¹⁰

San Diego, USA

Oral presentation

Dec 2025

Impact of vaccine-induced severity attenuation on bias in vaccine-effectiveness estimates from test-negative designs.

EPIDEMICS⁹

Bologna, Italy

Poster presentation

Nov 2023

Estimating the COVID-19 severity pyramid in Hong Kong, January 2020–January 2023.

HKCCM Annual Scientific Meeting

Hong Kong, China

Poster presentation

Sep 2022

Surveillance for local and importation risk of COVID-19.

RESEARCH EXPERIENCE

Laboratory of Data Discovery for Health (D²4H) Hong Kong, China
Research Associate Jul 2021–Jul 2022

- Supported data access, protocol development, and ethics approval for a Hospital Authority Data Collaboration Lab retrospective cohort study of oseltamivir effectiveness in hospitalized influenza patients using longitudinal clinical records.
- Co-developed an R package integrating EpiEstim and virosolver to simulate epidemic trajectories, estimate time-varying reproduction numbers, and support real-time monitoring of local COVID-19 transmission.
- Conducted systematic reviews and meta-analyses of SARS-CoV-2 transmissibility, synthesizing variant-specific reproduction numbers and superspreading parameters.

Southern University of Science and Technology Shenzhen, China
Research Assistant, Department of Statistics and Data Science Jul 2020–Jun 2021

- Conducted methodological literature reviews and developed estimation methods using the MM algorithm for bivariate binomial distributions and multivariate zero-inflated count data.

TEACHING EXPERIENCE

The University of Hong Kong
CMED 6020 Advanced Statistical Methods I, *Tutor* Spring 2025–Spring 2026
CMED 6020 Advanced Statistical Methods I, *Teaching Assistant* Spring 2023–Spring 2024

Southern University of Science and Technology
MA103A Linear Algebra A, *Teaching Assistant* Fall 2019
MA102B Calculus A II, *Teaching Assistant* Spring 2019
MA101B Calculus A I, *Teaching Assistant* Fall 2018

ADDITIONAL TRAINING

Summer Institute in Statistics and Modeling in Infectious Diseases (SISMID)
University of Washington, Department of Biostatistics July 2023
Selected modules: MCMC I and II for Infectious Diseases; Simulation-Based Inference for Epidemiological Dynamics; Statistics and Modeling with Novel Data Streams.

TECHNICAL SKILLS

Programming: R, Python; **Tools:** L^AT_EX, Git.

LANGUAGES

English (professional); Mandarin (native); Cantonese (conversational).